

- 8 (a) probability of decay of a nucleus M1
per unit time A1 [2]
- (b) $A = \lambda N$...(ignore sign)..... B1 [1]
- (c) (i) 1 m^3 contains $1 / 0.024 = 41.7 \text{ mol}$ C1
 1 m^3 contains $41.7 \times N_A = 2.5 \times 10^{25}$ molecules A1
(ii) number = $(2.5 \times 10^{25}) / (1.5 \times 10^{21}) = 1.67 \times 10^4$ A1
(iii) $\lambda T_{1/2} = 0.693$
 $\lambda = 0.693 / 56 = 0.0124 \text{ s}^{-1}$ C1
activity = $0.0124 \times 1.67 \times 10^4$
= 210 Bq A1 [5]